

Marco Alfredo Loaiza Carrillo

Distributed Systems & Edge-Cloud Engineer | Python Developer | PhD ICT

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PROFESSIONAL SUMMARY

Distributed systems, IoT, and Edge-Cloud engineer with 7+ years of experience and a PhD in ICT. Proven track record in ICT infrastructure and systems integration, including contributions to an EU Horizon Europe research project. Specialized in Kubernetes microservices, edge-cloud architectures, CI/CD with GitHub Actions, infrastructure automation with Terraform and Ansible, observability, and AI/ML-based systems.

TECHNICAL SKILLS

Infrastructure & Cloud: Kubernetes (K3s), Docker, Edge-Cloud Computing, Distributed Systems, Linux, Raspberry Pi

Programming & IoT: Python, C, C++, REST APIs, Microservices, MQTT, CoAP, Redis, Stream Processing, ESP32

DevOps & Observability: CI/CD (GitHub Actions), Terraform, Ansible, Prometheus, Grafana, OpenTelemetry, Power BI

AI/ML: AI/ML-based systems, data-driven analysis, multi-agent frameworks (SPADE), research and experimentation

PROFESSIONAL EXPERIENCE

Research Fellow

Dec 2025 – May 2026

DIMES — Università della Calabria, Rende, Italy

- Led the technical closure of the MLSysOps EU Horizon Europe project, coordinating final deliverables including consortium presentations, demo videos, and reports for the European Commission.
- Conducted experimental validation of the SPADE multi-agent framework in edge-cloud continuum scenarios, generating results for two ongoing research publications (a journal article and a workshop paper).

Research Engineer

Jan 2022 – Feb 2023

Corporacion Autonoma Regional de Cundinamarca — Bogota, Colombia

- Designed and developed the R&D technology platform for an Environmental Innovation Laboratory.
- Led competitive intelligence analysis producing Power BI dashboards to identify emerging research trends and new business opportunities for the organization.
- Contributed to technical documentation and institutional reports for internal stakeholders and external project partners.

Integration Engineer

Aug 2021 – Nov 2021

Whale Cloud — Bogota, Colombia

- Supported end-to-end delivery of a large-scale BSS project for WOM Telecomunicaciones, coordinating QA, development, and client teams across multiple delivery milestones.
- Monitored and validated e-commerce platform integration with the BSS core via REST APIs, ensuring SLA compliance and identifying issues before go-live.

Service Desk Analyst

Aug 2019 – Dec 2019

Schlumberger LTDA — Bogota, Colombia

- Provided IT support across 60+ countries, resolving complex incidents in O365, Active Directory, SCCM, data encryption, and VPN, with a high first-contact resolution rate.
- Facilitated a Design Thinking workshop for technical leadership to define and align strategic objectives for 2020.

Operating Systems Administrator

Dec 2016 – May 2018

IBM Colombia — Bogota, Colombia

- Managed infrastructure for 800+ servers serving major Colombian enterprise clients, maintaining high availability and strict SLA compliance.
- Served as Colombian liaison for a remote engineering team in India, driving automation initiatives that reduced manual workload and improved incident response times.

RESEARCH PROJECT

MLSysOps — EU Horizon Europe Project

Jan 2023 – Jan 2026

Universita della Calabria — Doctoral Research

- Designed and implemented agentic solutions using the SPADE multi-agent framework across distributed layers of the computing continuum, contributing to an ML-based IoT platform orchestrated with Kubernetes within a multi-partner EU consortium.
- Implemented a full observability stack (Prometheus, Grafana, OpenTelemetry) and automated infrastructure provisioning with Terraform and Ansible.
- Contributed to technical documentation, institutional reports, and 3 peer-reviewed publications (IEEE INISTA, EWSN, WOA) presented to the European Commission.

KEY PROJECTS

K8s GitOps Lab — Zero-Touch CI/CD Pipeline with Kubernetes 2024

Personal Project | *GitHub Actions · ArgoCD · k3s · Docker · GHCR · Python* | github.com/marcolo-30/k8s-gitops-lab

- Designed and implemented an end-to-end GitOps pipeline reducing deployment cycle from 20 minutes to 90 seconds, from git push to pod running in Kubernetes with zero manual intervention.
- Configured GitHub Actions as CI stage for multi-platform Docker image builds (AMD64/ARM64) and deployed ArgoCD on k3s as the CD stage.

IoT-Cloud Pipeline for ML-based Health Monitoring 2021

Personal Project | *ESP32 · AWS IoT Core · MQTT · Python · ML Pipeline*

- Designed an ESP32-based wearable device with SpO2 and heart rate sensors, transmitting data in real time to AWS IoT Core.
- Architected the AWS data pipeline as an entry point for Machine Learning models, enabling predictive analytics over physiological signals.

IoT Fleet Traceability System for Supply Chain 2022

MSc Thesis Project | *ESP32 · GPS · DHT · MQTT · ThingsBoard*

- Designed and implemented an end-to-end IoT node for fleet traceability: PCB design, component soldering, and hardware validation with GPS, temperature, and humidity sensors.
- Developed C/C++ firmware for real-time data transmission via MQTT to ThingsBoard, enabling GPS position and environmental variable visualization.

EDUCATION

PhD in Information and Communication Technologies — *Universita della Calabria, Italy* January 2026

MSc in Internet of Things Engineering — *Pontificia Universidad Javeriana, Colombia* September 2022

Electronic Engineering — *Pontificia Universidad Javeriana, Colombia* March 2017

PUBLICATIONS

- Bouleanu, D. C., Carrillo, M. A. L., et al. (2024). Benefits of Agent-Oriented Transitioning from Monolithic To Service-Based Architectures. IEEE INISTA 2024. DOI: 10.1109/INISTA62901.2024.10683829
- Loaiza, M., Savaglio, C., Arijó, N. H., et al. (2023). Agents in Software Development Architectures. WOA 2023, CEUR-WS.org.
- Loaiza, M., Savaglio, C., Gravina, R., et al. (2023). Agents in the Computing Continuum: the MLSysOps Perspective. EWSN 2023. DOI: 10.5555/3639940.3639991
- Loaiza, M. (2022). IoT Solution for Product Traceability in the Transport Stage of a Supply Chain. M.Sc. Thesis, Pontificia Universidad Javeriana.

LANGUAGES

Spanish (Native) | English (Professional) | Italian (Basic)